

Candidate Data Transformer

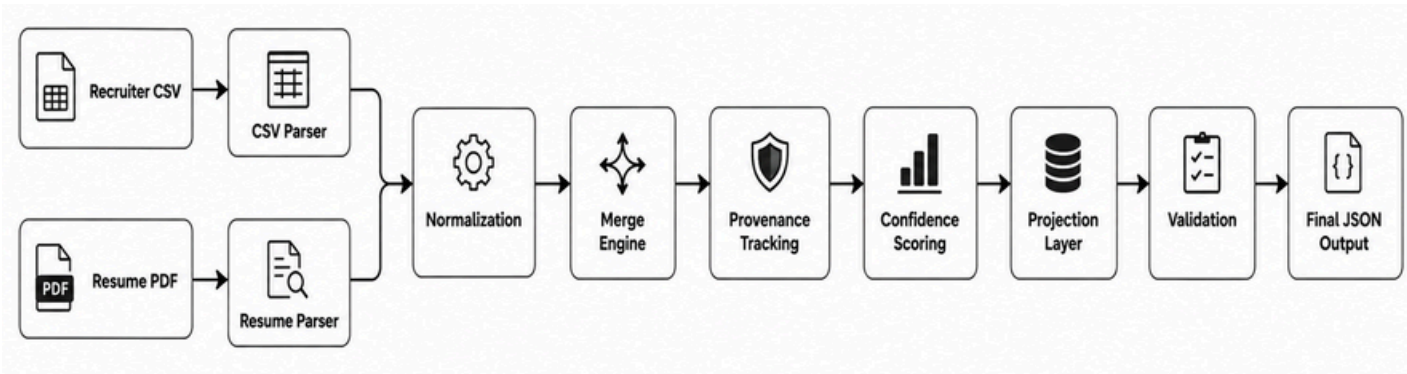
Technical Design Document

Eightfold AI Assignment | Aashish Poddar

Problem Statement

Design a configurable data transformation pipeline that consolidates structured (Recruiter CSV) and unstructured (Resume PDF) candidate information into a unified canonical JSON profile through parsing, normalization, conflict resolution, provenance tracking, confidence scoring, validation, and configurable output generation.

Processing Pipeline



Canonical Schema

```
{
  "full_name": "",
  "email": "",
  "phone": "",
  "current_company": "",
  "title": "",
  "provenance": {},
  "confidence": {}
}
```



Merge Policy

- Resume preferred for Name
- Resume preferred for Email
- Resume preferred for Phone
- Recruiter CSV provides Company
- Recruiter CSV provides Title
- Missing values filled from available sources



Confidence

Source	Score
Resume PDF	0.95
Recruiter CSV	0.90



Validation & Projection

Validation

- ☒ Name required
- ☒ Email required
- ☒ Phone required

Projection

- default_config.json
 - custom_config.json
- Runtime configuration controls the output fields.



Edge Cases

- Missing email
- Missing phone
- Empty recruiter CSV
- Invalid Resume PDF
- Missing configuration file



Future Improvements

- LinkedIn API Integration
- GitHub API Integration
- AI-based Skill Extraction
- Dynamic Confidence Scoring
- Duplicate Candidate Detection